

AN ALGEBRAIC PROOF OF GHOSH'S LEMMA 3.7: INTERSECTION THEORY IN PLACE OF BROUWER DEGREE

ABSTRACT. In his proof of the Casas–Alvero conjecture, Ghosh [3] establishes an existence lemma for “almost counterexamples” (his Lemma 3.7) by means of Abel–Gontcharoff polynomials and the topological Brouwer degree over $\mathbb{C}$. This is the only analytic ingredient of the paper, and its proof occupies pages 11–13. We observe that, in the setting where the lemma is actually applied —inside the downward inductive step, where the conjecture is already assumed in degree $n + 1$ — the statement follows from the projective dimension theorem in a few lines. The resulting proof is purely algebraic, is valid over any algebraically closed field with no hypothesis on the characteristic, and removes every dependence on the Euclidean topology.

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1. NOTATION

We follow [3]. Let $\mathbb{K}$ be algebraically closed, $n \geq 2$, and $R_n = \mathbb{K}[x_1, \dots, x_n]$. For $1 \leq j \leq n+1$ let $\Phi_j: R_n \rightarrow R_n$ be the $\mathbb{K}$ -algebra automorphism

$$\Phi_j(x_l) = x_l - x_j \ (l \neq j), \quad \Phi_j(x_j) = -x_j, \quad \Phi_{n+1} = \text{id},$$

and let e_k denote the elementary symmetric polynomial of degree k in $x_1, \dots, x_n$. For $1 \leq i \leq n$ and $1 \leq j \leq n + 1$ put

$$F(i, j) := \Phi_j(e_{n+1-i}), \quad \deg F(i, j) = n + 1 - i.$$

These forms encode, for $f(X) = X \prod_{i=1}^n (X - x_i)$ of degree $n + 1$, the condition that f and its i -th Hasse derivative share the root x_j : up to sign, $F(i, j)$ is $f_i(x_j)$ written in the roots. For a fixed choice of indices $\mathbf{j} = (j_1, \dots, j_n)$ we write $F_i := F(i, j_i)$.

We use the reformulation [3, Prop. 1.4] (from [2]):

Proposition 1.1 ([3]). *The Casas–Alvero conjecture holds in degree $n + 1$ over $\mathbb{K}$ if and only if for every choice $\mathbf{j}$ the sequence $F_1, \dots, F_n$ is a regular sequence in R_n .*

Since the F_i are n forms in n variables, being a regular sequence is equivalent to being a system of parameters, that is,

$$V(F_1, \dots, F_n) = \{0\} \subseteq \mathbb{A}^n \quad \Longleftrightarrow \quad V_+(F_1, \dots, F_n) = \emptyset \subseteq \mathbb{P}^{n-1}. \quad (\text{H})$$

2. THE LEMMA

The statement needed in [3] (his Lemma 3.7, via the Nullstellensatz) is: for each l there exists a degree $n + 1$ polynomial f with $f_i(x_{j_i}) = 0$ for all $i > l$ but $f_l(x_{j_l}) \neq 0$; equivalently,

$$F_l \notin \sqrt{(F_{l+1}, \dots, F_n)}.$$

Lemma 2.1. *Let $\mathbb{K}$ be algebraically closed and assume (H) holds for the choice $\mathbf{j}$. Then, for every $1 \leq l \leq n$,*

$$F_l \notin \sqrt{(F_{l+1}, \dots, F_n)} \quad \text{in } R_n.$$

Proof. Assume the contrary and set $Z := V_+(F_{l+1}, \dots, F_n) \subseteq \mathbb{P}^{n-1}$.

The F_i are nonzero forms (each is a linear change of variables applied to $e_{n+1-i} \neq 0$), so Z is cut out by $n - l \leq n - 1$ hypersurfaces of $\mathbb{P}^{n-1}$. By the projective dimension theorem, $Z \neq \emptyset$ and every irreducible component of Z satisfies

$$\dim Z \geq (n - 1) - (n - l) = l - 1.$$

By assumption F_l vanishes on Z , i.e. $Z \subseteq V_+(F_l)$, and therefore

$$V_+(F_l, F_{l+1}, \dots, F_n) = Z, \quad \dim V_+(F_l, \dots, F_n) \geq l - 1.$$

Now cut with the remaining $l - 1$ hypersurfaces $V_+(F_1), \dots, V_+(F_{l-1})$. Each intersection with a hypersurface of $\mathbb{P}^{n-1}$ drops the dimension by at most 1, and is nonempty as long as the current dimension is ≥ 1 ; performing the $l - 1$ cuts starting from dimension $\geq l - 1$ yields

$$\dim V_+(F_1, \dots, F_n) \geq (l - 1) - (l - 1) = 0,$$

so in particular $V_+(F_1, \dots, F_n) \neq \emptyset$, contradicting (H). $\square$

Remark 2.2. The proof uses nothing about the characteristic of $\mathbb{K}$: only Krull's dimension theorem and the fact that the F_i are nonzero forms. In particular,

$$\text{CA in degree } n + 1 \text{ over } \mathbb{K} \implies \text{Lemma 3.7 of [3] over } \mathbb{K},$$

in any characteristic. The version in [3] is unconditional but is proved only over $\mathbb{C}$.

Remark 2.3. A point of $V_+(F_1, \dots, F_n)$ is precisely a counterexample to Casas–Alvero in degree $n + 1$ whose recycled roots are prescribed by $\mathbf{j}$. Informally, the proof says that if the l -th datum were a consequence of the later ones, then the full system would be overdetermined by one equation fewer, and would therefore admit a nontrivial solution.

3. WHY THE CONDITIONAL VERSION SUFFICES

Lemma 3.7 of [3] is used only in the proof of his Theorem 3.6 (that $\mu_{R_n}(\mathcal{I}_n(\mathbf{j})) = n$), and the latter only through Corollary 3.9. In turn, Corollary 3.9 is invoked twice, in the proofs of Proposition 4.3 and of Lemma 4.5, both inside Section 4.2, that is, under the inductive hypothesis 4.1.1: *the conjecture holds in degree $n + 1$* . That hypothesis is exactly (H).

Consequently, Lemma 2.1 can replace Lemma 3.7 without altering the structure of the argument: it suffices to state Theorem 3.6 in its conditional form (“under (H), $\mu_{R_n}(\mathcal{I}_n(\mathbf{j})) = n$ ”). Note also that the form required in [3, (3.9)] is non-membership in the *ideal* $(F_i : i > l)$, which is weaker than the non-membership in the radical proved here.

Remark 3.1. We would suggest that the author double-check this usage point: the reading above comes from tracing the cross-references of the preprint (v2, March 2026), not from a full rewriting of the paper.

4. WHAT IS REMOVED AND WHAT IS GAINED

- The Abel–Gontcharoff polynomials, the incidence variety $\mathcal{V}$, the space $\mathbb{P}(\mathcal{L})$ of linear surjections, the paths γ , the balls $B_R(\mathbf{0})$ and spheres $S_{\epsilon_i}(z_i)$, the homotopy invariance of the Brouwer degree and the positivity of the local holomorphic index ([1, Part I, §5.4]) all disappear. These are essentially pages 11–13 of [3].

- The only analytic dependence disappears: the argument becomes valid over any algebraically closed field, not just over $\mathbb{C}$, and in any characteristic.
- Remarks 3.8(1) and 3.8(2) of [3], which handle special cases by explicit witnesses and by Rolle's theorem, become subsumed.

The moral is the usual one in intersection theory: the positivity of the local index of a holomorphic map, which [3] uses to guarantee that the degree does not vanish, is unnecessary if one argues with dimensions in projective space, where the nonemptiness of an intersection of few hypersurfaces is automatic.

5. COMPUTATIONAL VERIFICATION

Using SageMath 10.7, in the root-space formulation, we verified for all choices $\mathbf{j}$ and all l :

	$n = 3$ ($d = 4$)	$n = 4$ ($d = 5$)
hypothesis (H): $\dim V(F_1, \dots, F_n) = 0$	64/64 branches	625/625 branches
conclusion: $F_l \notin \sqrt{(F_i : i > l)}$	84/84 cases	780/780 cases
dimension bound	sharp ($= l$)	sharp ($= l$)
$\dim V(F_{l+1}, \dots, F_n) \geq l$		
cases covered by [3, Rmk. 3.8(1)]	43%	23%

The observed dimensions are exactly l , the minimum allowed by the dimension theorem: the argument rests on the tight case. The last row also shows that the proportion of cases that [3] settles by explicit witnesses decreases with the degree, so the step replaced here is the one carrying almost all of the work.

REFERENCES

- [1] V. I. Arnold, S. M. Gusein-Zade, A. N. Varchenko, *Singularities of Differentiable Maps, Volume 1*, Birkhäuser, 2012.
- [2] S. Ghosh, *A finiteness result towards the Casas-Alvero conjecture*, arXiv:2402.18717.
- [3] S. Ghosh, *Proof of the Casas-Alvero conjecture*, arXiv:2501.09272 (v2, March 2026).